Christof Teuscher

ECE 410/510: Hardware for AI and ML, Spring 2026

Week 4: Recap, outlook, and reminders

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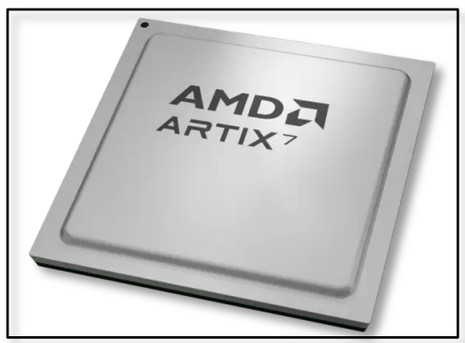

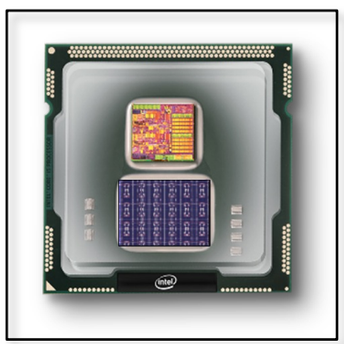

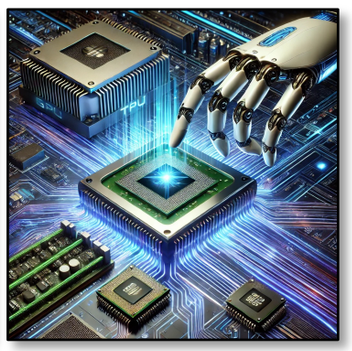

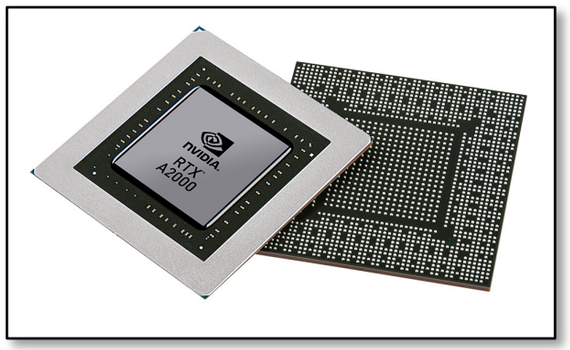

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What did you learn last week?

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www.teuscher-lab.com/teaching

Claudian Your friendly ECE 410/510 Hardware for AI and ML TA!

teuscher@pdx.edu Sign out

Codefest 3 Time remaining: **9:34**

Where do I start with CMAN?

Welcome! Great question to start with. CMAN is all about analyzing DRAM traffic for matrix multiplication, so let's make sure you have a solid foundation before diving into the calculations.

Here are a few warm-up questions to get you oriented:

- **What is the size of the matrices?** The problem says N×N where N=32. Each element is FP32 — how many bytes is that per element?
- **Think about the naive triple loop.** For a single output element C[i][j], you compute a dot product. How many multiplications and additions does that require? Which elements of A and B do you need to read?
- **"No data reuse"** is a key assumption in Task 1. What does that mean in terms of where every single memory access goes?

Start by thinking through Task 1 — just one output element C[i][j] first — before worrying about the full N×N output.

What's your initial thinking on how many times a single element of B gets accessed across the whole computation? 😊

All conversations are logged and may be reviewed for course improvement and analytical purposes.

Type your question... (Enter to send, Shift+Enter for new line) End Send

14	10	725,917
Total sessions	Registered students	Total tokens used

Now:

- 2 sessions per day
- 15min per session

https://teuscher-lab.ece.pdx.edu/claudian

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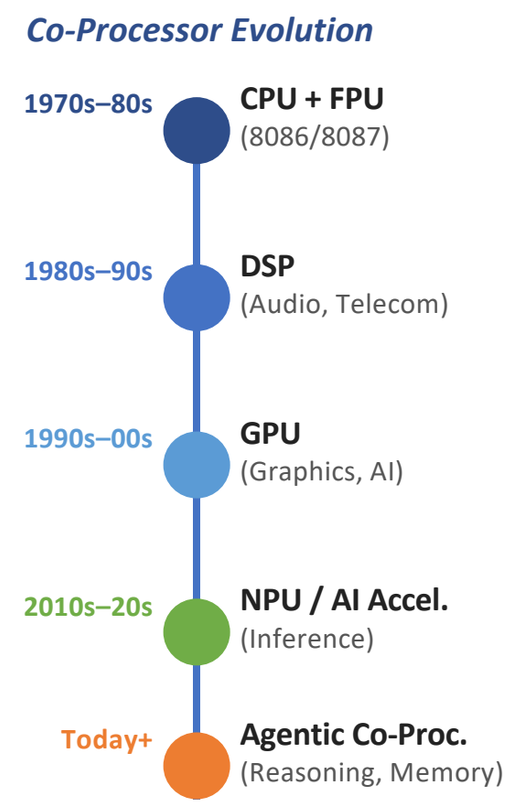
A new era for co-processing

- **AI is driving a new era of co-processor specialization**
- Hardware architectures are evolving faster than design, verification, and production cycles can keep up.
- Workloads shifting from short kernel-style inference to long-running agentic loops requiring memory, tool use, and reasoning.
- Winning co-processors minimize data movement, software friction, and verification risk simultaneously.
- NPUs evolving beyond fixed-function MAC engines toward flexible, programmable architectures.
- Full programmability is what determines which new architectures ultimately succeed



https://semiengineering.com/a-new-era-for-co-processing

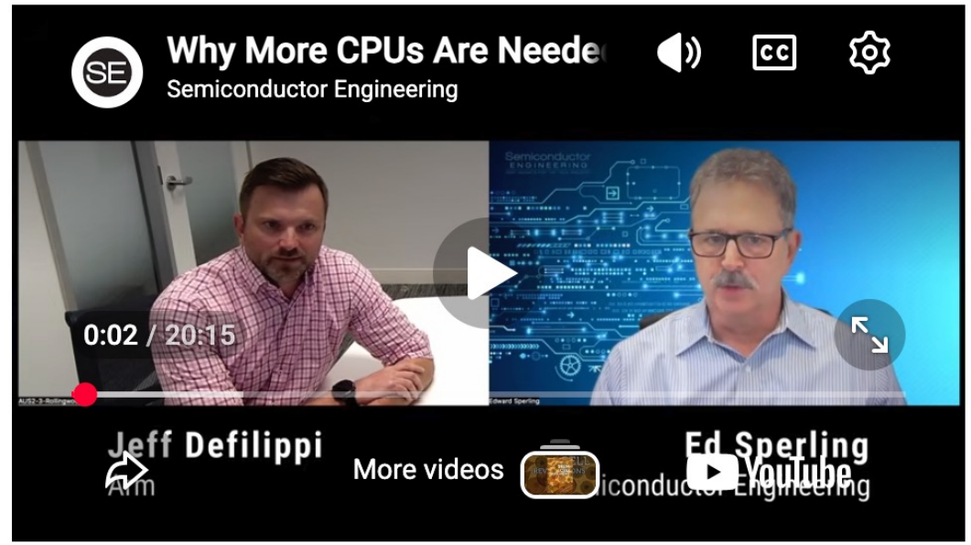
Why More CPUs Are Needed For Agentic AI

General-purpose processing demands will multiply once machines are talking to machines.

APRIL 15TH, 2026 - BY: **ED SPERLING**

The shift from generative AI to agentic AI will significantly increase the amount of compute power needed in data centers. Queries to search for and analyze data from multiple sources will be performed simultaneously by agents and without human intervention, rather than a single request from a live person. Jeff Defilippi, senior director of product management at Arm, talks about the impact of running these queries 24 x 7, how queries will be orchestrated across agents, how compute resources need to be architected to handle all of this, and how to improve the bandwidth in the chip architecture to allow for faster data movement.

SEMICONDUCTOR ENGINEERING
DEEP INSIGHTS FOR THE TECH INDUSTRY



https://www.youtube.com/watch?v=cInO-qTGS1A

https://semiengineering.com/why-more-cpus-are-needed-for-agentic-ai

NVIDIA is currently the only choice for AI

But that is changing. Relying on a single vendor is a massive vulnerability.

Chip design as we know it is over

Computer Architecture's AlphaZero Moment: Automated Discovery in an Encircled World

Karthikeyan Sankaralingam
NVIDIA Research
karus@nvidia.com

April 7, 2026

Reading for this week!

https://arxiv.org/abs/2604.03312

We're entering a world where:

- LLMs can generate hundreds of viable architectural ideas per day.
- Automated pipelines can evaluate thousands of designs per week.
- Design cycles are shortened from months to weeks.

Evaluation is the new bottleneck.

- "[…] human-driven architecture research is fundamentally ill-suited for this new era. The architectural design space is vast […], yet human teams explore perhaps 50-100 designs per generation, sampling less than 0.001% of possibilities."
- "We predict that within 2 years, purely human-driven architecture research will be as obsolete as human chess players competing against engines."

This week

Today

- Some more GPU + CUDA
- Numerical precision: trading throughput for precision
- Mapping deep neural nets (CNN) on GPUs
- VLSI design flow + tools (project M2 prep)

Wednesday

- CMAN: numerical precision
- CLLM: INT8 MAC unit in Verilog
- COPT: TBA

Project milestone 2 (M2)

- First working simulation:
  - HDL description of the compute core with a testbench demonstrating correct output for a representative input.
  - Interface module in HDL with testbench exercising at least one write and one read/response transaction.
  - Precision choice documented (FP32 / FP16 / BF16 / INT8 / other) with quantization error analysis if applicable.

- Due: **Sun, May 3**

Next week: quiz 1

How to best prepare?

- Attend the lectures and the codefests
- Complete the codefest challenges
- Digest the materials
- Generate a study guide with your LLM
- Generate sample questions with your LLM
- Generate a cheat sheet

Modality:

- TBA